

The v4 Rigor Index — Formal Specification

MSDS 498 Capstone · July 2026 · Code: `etl/build_rigor_classification.py` (v4 component definitions)

Layer 0 — Derived inputs

AP qualifying density (the v4 AP-performance metric). With t_i = AP tests taken per student and $\bar{s}_i$ = mean AP exam score for school i :

$$QD_i = t_i \cdot \Pr(\text{score} \geq 3) = t_i \cdot \Phi\left(\frac{\bar{s}_i - 2.5}{1.2}\right)$$

where Φ is the standard normal CDF and 1.2 is the assumed within-school score SD (a documented approximation; the College Board does not publish school-level score distributions). Winsorized at the 99th percentile.

IB intensity (the verified IB signal). With IB_i^{enr} = CRDC IB enrollment, E_i = grade 9–12 enrollment, and $F_i \in \{0, 1\}$ the verified IB flag:

$$IBint_i = \begin{cases} IB_i^{enr}/E_i & \text{if CRDC enrollment is available} \\ F_i & \text{otherwise} \end{cases}$$

Winsorized at the 99th percentile.

Layer 1 — Standardization

Each raw sub-feature x is z -scored over all schools reporting it:

$$z(x_i) = \frac{x_i - \mu_x}{\sigma_x}$$

Layer 2 — Component scores

Each component score C_{ik} is the mean of its z -scored sub-features, taken **only over the sub-features school i actually reports** (no imputation):

Component k	Sub-features (each z -scored)	Weight w_k
AP opportunity	AP tests taken; AP classes offered; AP take rate	0.25
AP performance	AP qualifying density QD	0.20
CRDC coursework	AP participation; dual-enrollment rate; IB intensity IBint	0.20
Test participation	SAT/ACT test-taker rate; SAT participation	0.15
Test performance	mean SAT; ACT composite (IL)	0.20

The two entries in bold, plus the IBint term, are the v4 revisions; a standalone IB component remains at weight 0.

Layer 3 — Weighted composite with proportional reallocation

Let $A_i \subseteq \{1, \dots, 5\}$ be the set of components available for school i . The rigor score is

$$R_i = \frac{\sum_{k \in A_i} w_k C_{ik}}{\sum_{k \in A_i} w_k}$$

i.e., weights reallocate proportionally over whatever the school reports. If $A_i = \emptyset$, the school receives **no score** — it is logged as unscored, never defaulted to a middle tier.

Layer 4 — Tier assignment

Tiers are cut by Jenks natural breaks, computed as one-dimensional k -means on $\{R_i\}$ with $k = 5$ (fixed seed). Cluster centroids are ordered low $\rightarrow$ high and mapped onto:

Below Average $\rightarrow$ Average $\rightarrow$ Demanding $\rightarrow$ Very Demanding $\rightarrow$ Most Demanding

An equal-frequency quintile assignment is emitted alongside as the cut-point sensitivity check (the two schemes agree for 40.8% of scored schools — reported, not hidden).

Reported diagnostics (same script)

Nominal vs. effective weights (variance decomposition on the full-coverage subset); sensitivity table over alternate weighting schemes (Spearman rank correlation and % of schools changing tier); CRDC-loss scenario; and the poverty/funding correlation of the resulting tiers ($\rho = -0.11$ with county child poverty in v4).